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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester B.Tech (CBCS) Degree Examinations

December 2024

MA1005-1 – MATRIX ALGEBRA AND DIFFERENTIAL EQUATIONS

(For BT, CV, ME)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **One full** question from each **Unit**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Rank of the matrix $A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 4 & 2 & 3 & 0 \\ 1 & 0 & 0 & 0 \\ -8 & 0 & 0 & 0 \end{bmatrix}$ is
A) 1
B) 2
C) 3
D) 4
2. The eigen values of $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ are
A) 2,3
B) 2,2
C) -2,3
D) -3,-2
3. The characteristic equation of the matrix A is
A) $|A - \lambda I|X = 0$
B) $|A - \lambda I| = I$
C) $|A + \lambda I| = 0$
D) $|A - \lambda I| = 0$
4. The solution of the system of equations
 $2x + y = 1, x + z = 3, y + z = 1$
A) $x = -1, y = -1, z = 2$
B) $x = -1, y = 1, z = 2$
C) $x = 1, y = -1, z = 2$
D) $x = 1, y = -1, z = -2$
5. The p-series is convergent for
A) $p > 1$
B) $p < 1$
C) $p \leq 1$
D) $p \geq 1$
6. If $\sum u_n$ is convergent series of positive terms, then $\lim_{n \rightarrow \infty} u_n$ is ____
A) ∞
B) 1
C) 0
D) -1
7. The Maclaurin's series expansion of e^x is ____
A) $1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots \infty$
B) $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \infty$
C) $x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \infty$
D) $x - \frac{x^2}{2!} + \frac{x^3}{3!} - \dots \infty$
8. In a positive term series $\sum u_n$ if $\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lambda$ then the series diverges for
A) $\lambda \leq -1$
B) $\lambda < 1$
C) $\lambda = 1$
D) $\lambda > 1$

9. The order and degree of the differential equation $2x \frac{d^4y}{dx^4} + 5x^2 \left(\frac{dy}{dx}\right)^3 - xy = 0$ respectively are
A) 3, 1
B) 4, 1
C) 3, 4
D) 1, 4
10. The differential equation $p = \sin(px - y)$ is, where $p = \frac{dy}{dx}$
A) Clairaut's equation
B) Bernoulli's equation
C) Linear equation
D) Exact equation
11. The differential equation $y - 2px = \tan^{-1}(xp^2)$ is _____
A) linear
B) solvable for $p = \frac{dy}{dx}$
C) solvable for x
D) solvable for y
12. In order to find the differential equation of orthogonal trajectories of the curves $f(x, y, c) = 0$, which of the following replacements is made
A) $\frac{dy}{dx}$ is replaced by $\frac{dx}{dy}$
B) $\frac{dy}{dx}$ is replaced by $-\frac{dx}{dy}$
C) $\frac{y}{x}$ is replaced by $\frac{x}{y}$
D) $\frac{x}{y}$ is replaced by $-\frac{x}{y}$
13. General solution of the differential equation $(D^2 + 12D + 36)y = 0$ is
A) $y = (c_1 + c_2x)e^{-6x}$
B) $y = (c_1 + c_2x)e^{-5x}$
C) $y = (c_1 + c_2x)e^{5x}$
D) $y = (c_1 + c_2)e^{-6x}$
14. The particular integral of the differential equation $(D - 3)^2y = e^{3x}$ is
A) $\frac{x}{2}e^{3x}$
B) $\frac{e^{3x}}{2}$
C) $\frac{x^2}{2}e^{3x}$
D) $-\frac{x^2}{6}e^{3x}$
15. The linear differential equation with constant coefficients corresponding to the equation $x^2 \frac{d^2y}{dx^2} - x \frac{dy}{dx} + y = \log x$ is
A) $(D + 1)^2y = t$
B) $(D - 1)^2y = t$
C) $(D + 1)^2y = x$
D) $(D - 1)^2y = x$
16. If D is the differential operator, then for any function $f(x)$, $\frac{1}{D}f(x)$ represents:
A) $\frac{f(D)}{D}$
B) $\int f(D)dx$
C) $\frac{d}{dx}f(x)$
D) $\int f(x)dx$
17. Which of the following is not a definition of Gamma function?
A) $\Gamma(n) = n!$
B) $\Gamma(n) = \int_0^\infty x^{n-1}e^{-x}dx$
C) $\Gamma(n + 1) = n\Gamma(n)$
D) $\Gamma(n) = \int_0^1 \log\left(\frac{1}{u}\right)du$
18. The value of $\Gamma\left(\frac{1}{2}\right)$?
A) $\sqrt{\pi}$
B) $\left(\sqrt{\frac{\pi}{2}}\right)$
C) $\frac{\sqrt{\pi}}{2}$
D) $\frac{\pi}{2}$
19. Which of the following formula can be used to find area of a plane region D in polar form?
A) $A = \iint dr d\theta$
B) $A = \iint r dr d\theta$
C) $A = \iint f(r, \theta) dr d\theta$
D) $A = \iint D dr d\theta$
20. Value of $\int_0^3 \int_1^2 \int_0^2 5 dx dy dz$ is
A) 30
B) 6
C) 5
D) 60

PART - B: DESCRIPTIVE ANSWER QUESTIONS

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	Find the eigen values of $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$	4	L*2	1	1
	b)	Define Rank of the matrix. Find the rank of the following matrix using elementary row transformation. $\begin{bmatrix} 1 & 3 & 1 & -2 & -3 \\ 1 & 4 & 3 & -1 & -4 \\ 2 & 3 & -4 & -7 & -3 \\ 3 & 8 & 1 & -7 & -8 \end{bmatrix}$	6	L2	1	1
	c)	Using power method find the largest eigen value and corresponding eigen vector of $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ by taking $X^0 = [1 \ 0 \ 0]^T$ carry out 5 iterations.	6	L2	1	1
2.	a)	Test for consistency of the system of linear equations $x - 4y + 5z = 8$ $3x + 7y - z = -3$ $x + 15y - 11z = -14$	4	L2	1	1
	b)	Using Gauss seidel method solve the system of equations $10x + y + z = 12$ $x + 10y + z = 12$ $x + y + 10z = 12$ take $x^0 = y^0 = z^0 = 0$, carry out three iterations.	6	L2	1	1
	c)	Find the eigen values and corresponding eigen vectors of the matrix $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$	6	L3	1	1
		Unit – II				
3.	a)	Test for convergence of $\sum_{n=1}^{\infty} 3^n \left(\frac{n}{n+1}\right)^{n^2}$	4	L1	2	1
	b)	Define D Alembert's Ratio test and test for convergence of $\left(\frac{1}{3}\right)^2 + \left(\frac{1.2}{3.5}\right)^2 + \left(\frac{1.2.3}{3.5.7}\right)^2 + \dots \dots \dots \infty$	6	L2	2	1
	c)	Obtain Taylor's series expansion of $\log x$ about $x = 1$ upto three non-vanishing terms.	6	L2	2	1
4.	a)	Test for convergence of the series $\sum \frac{n^2+6}{5n^5+8}$	4	L1	2	1
	b)	Test for convergence of $\frac{1}{2\sqrt{1}} + \frac{x^2}{3\sqrt{2}} + \frac{x^4}{4\sqrt{3}} + \dots \dots \dots \infty$; $x > 0$	6	L2	2	1
	c)	Obtain the Maclaurin's expansion for $\tan x$ upto 4 th degree term	6	L2	2	1
		Unit – III				
5.	a)	Obtain the general and singular solution of $\sin(px - y) = p$	4	L2	3	1
	b)	If a body originally is at 80°C cools down to 60°C in 20 minutes, the temperature of air being 40°C. Find the temperature of the body after 40 min from the original.	6	L3	3	1
	c)	Find the orthogonal trajectories of the family of curves $r = a(1 - \cos\theta)$	6	L2	3	1
6.	a)	Solve the Differential equation $\frac{dy}{dx} + y \tan x = y^3 \sec x$	4	L1	3	1
	b)	Solve $y + px = x^4 p^2$	6	L2	3	1
	c)	Solve $p^3 + 2xp^2 - y^2p^2 - 2xy^2p = 0$	6	L2	3	1

Unit – IV

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|-------|---|---|----|---|---|
| 7. a) | Solve the Differential equation $(D^3 - 6D^2 + 11D - 6)y = 0$ | 4 | L1 | 4 | 1 |
| b) | Find the solution of $(D^2 - 4D + 3)y = \sin 3x \cos 2x$ | 6 | L2 | 4 | 1 |
| c) | Using method of variation of parameters solve
$(D^2 - 4D + 4)y = \frac{e^{2x}}{x}$ | 6 | L2 | 4 | 1 |
| 8. a) | Solve the Differential Equation $(D + 2)^2 y = e^{2x}$ | 4 | L1 | 4 | 1 |
| b) | Find the solution of $(D^3 - D)y = 2x + 2e^x$ | 6 | L2 | 4 | 1 |
| c) | Solve $x^2 y'' - 2y = x^2 + \frac{1}{x}$ using Cauchy's method | 6 | L2 | 4 | 1 |

Unit – V

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|--------|--|---|----|---|---|
| 9. a) | Evaluate $\int_0^1 \int_0^2 \int_1^2 x^2 yz \, dx \, dy \, dz$ | 4 | L1 | 5 | 1 |
| b) | Prove that $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$ | 6 | L2 | 5 | 2 |
| c) | Evaluate $I = \iint_R xy(x+y) \, dx \, dy$ where R is the region bounded by the curves $y = x^2$ & $y = x$ | 6 | L3 | 5 | 1 |
| 10. a) | Evaluate $I = \int_0^\infty x^4 e^{-x^2} \, dx$ using Gamma functions | 4 | L1 | 5 | 1 |
| b) | Using double integral, find the area of the cardioid
$r = a(1 + \cos \theta)$ | 6 | L3 | 5 | 2 |
| c) | Find the volume common to the cylinder $x^2 + y^2 = a^2$ &
$x^2 + z^2 = a^2$ | 6 | L3 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
